



BLOCKCHAIN

Complete Learning Roadmap

Understand distributed ledgers, smart contracts, and secure on-chain application design.

Ledger fundamentals

Consensus concepts

Wallet basics

Cryptographic primitives

Transaction lifecycle

Token standards

01

Fundamentals

02

Core Concepts

03

Tools and Frameworks

04

Real Projects

05

Advanced Topics

5

Learning Stages

20+

Core Topics

10+

Projects

100%

Blockchain



Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

Ledger fundamentals • Consensus concepts • Wallet basics • Cryptographic primitives

02

Core Concepts

Transaction lifecycle • Token standards • Contract security basics • Gas mechanics

03

Tools and Frameworks

Solidity

04

Real Projects

ERC-20 project • Simple DAO voting contract • Contract audit checklist • On-chain analytics dashboard

05

Advanced Topics

MEV awareness • Cross-chain interoperability • Formal verification basics • Protocol governance models

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.



Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Ledger fundamentals
- Consensus concepts
- Wallet basics
- Cryptographic primitives

Tools & Frameworks

- Ledger fundamentals
- Consensus concepts
- Wallet basics
- Cryptographic primitives

Recommended Projects

- Blockchain workshop using Ledger fundamentals
- Blockchain portfolio project with Consensus concepts

Stage 1 Milestone Complete Blockchain workshop using Ledger fundamentals and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- Transaction lifecycle
- Token standards
- Contract security basics
- Gas mechanics

Tools & Frameworks

- Transaction lifecycle
- Token standards
- Contract security basics
- Gas mechanics

Recommended Projects

- Blockchain workshop using Transaction lifecycle
- Blockchain portfolio project with Token standards

Stage 2 Milestone Complete Blockchain workshop using Transaction lifecycle and document the decisions, trade-offs, and validation checks before moving to Stage 3.

Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- Solidity
- Hardhat/Foundry
- On-chain indexing tools
- Wallet integration libraries

Tools & Frameworks

- Solidity

Recommended Projects

- Blockchain workshop using Solidity
- Blockchain portfolio project with Hardhat/Foundry

Stage 3 Milestone Complete Blockchain workshop using Solidity and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- ERC-20 project
- Simple DAO voting contract
- Contract audit checklist
- On-chain analytics dashboard

Tools & Frameworks

- ERC-20 project
- Simple DAO voting contract
- Contract audit checklist
- On-chain analytics dashboard

Recommended Projects

- Blockchain workshop using ERC-20 project
- Blockchain portfolio project with Simple DAO voting contract

Stage 4 Milestone Complete Blockchain workshop using ERC-20 project and document the decisions, trade-offs, and validation checks before moving to Stage 5.



Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

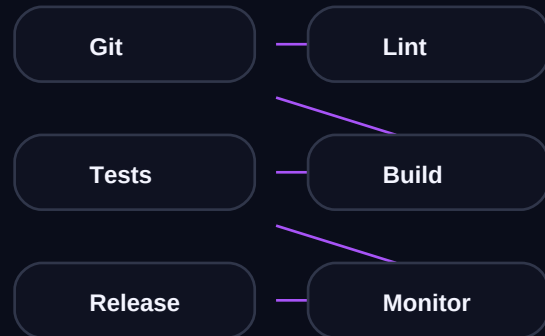
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- MEV awareness
- Cross-chain interoperability
- Formal verification basics
- Protocol governance models

CI/CD Pipeline



Stage 5 Milestone

A production-grade blockchain workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

Blockchain docs and community guides

Link: <https://ethereum.org/developers/docs/>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/blockchain>

PAID

Structured specialization track focused on Blockchain

Link: <https://roadmap.sh/blockchain>

FREE

Blockchain roadmap reference

Link: <https://roadmap.sh/blockchain>

FREE

Official documentation

Link: <https://ethereum.org/developers/docs/>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in Blockchain?
- Which project best proves Stage 4 competency in Blockchain?
- What trade-offs appear in Stage 5 decisions for Blockchain?